

GALLIO
or
The Tyranny of Science



By
J. W. N. SULLIVAN

GALLIO
or
The Tyranny of Science



By
J. W. N. SULLIVAN

The Project Gutenberg eBook of Gallio

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Gallio
or, The tyranny of science

Author: J. W. N. Sullivan

Release date: August 3, 2025 [eBook #76629]

Language: English

Original publication: New York: E. P. Dutton & Co., 1928

Other information and formats: www.gutenberg.org/ebooks/76629

Credits: Produced by Donald Cummings and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This book was produced from images made available by the HathiTrust Digital Library.)

*** START OF THE PROJECT GUTENBERG EBOOK GALLIO ***

GALLIO
OR
THE TYRANNY
OF
SCIENCE

GALLIO

OR

The Tyranny of Science

BY

J. W. N. SULLIVAN



E. P. DUTTON & CO. :: NEW YORK

GALLIO, OR THE TYRANNY OF SCIENCE
COPYRIGHT 1928 BY E. P. DUTTON & COMPANY
ALL RIGHTS RESERVED :: PRINTED IN U.S.A.

LIST OF SECTIONS

1
2
3
4
5

GALLIO

GALLIO

OR

THE TYRANNY OF SCIENCE

1

There can be no doubt that the prestige of science has greatly increased of recent times. In the days when Dickens wrote *The Mudfog Papers* the man of science, to the general reading public, was a purely comic figure. After the man of science had knocked the bottom out of the Victorian universe with his theory of Natural Selection he inspired the respect we accord to whatever is both powerful and sinister. He was observed, warily and acutely, as an enemy. This reaction was perfectly justified, for science, as expounded to the populace by such men as Huxley and Tyndall, deprived life of all that had hitherto made it worth living. The gravamen of their offence was not that they made man an integral part of the animal kingdom, but that they presented him with a universe that was entirely purposeless. Such a doctrine would probably come as a shock even to a disillusioned and emaciated Eastern Sage, but to the men of the Victorian age, almost every one of them brought up in an orthodox Christian household and filled with that belief in a wise Providence that comes of great material prosperity, it was nothing short of an outrage. Even the men of science themselves found their great discovery more than a little disconcerting. Nobody who reads them can fail to detect something strained, something occasionally almost frenzied, in their insistence on the duty of intellectual honesty. These men are, half the time, shouting aloud in order to hearten themselves. They were quite consciously martyrs to the truth. This is true, at any rate, of such men as Huxley and Clifford. There were many men of science, of course, who were not sufficiently alive to live in a universe of any description. Outside, their laboratories they had no perceptible existence. Many of them died simple Christians. But to all interested in such matters it became evident that the goal of science was the

detailed explanation of man as the accidental outcome of “matter and motion”. Since the arguments of the man of science could not be met (for only science can cast out science) the only thing left was to abuse him. This was magnificently done by Nietzsche, and rather less magnificently by Dostoevsky and Tolstoi. Nietzsche pointed out that the man of science was not a human being. He was merely an instrument, the most costly, the most exquisite, the most easily tarnished of instruments. He was incapable of love; he was incapable of hate. His one purpose was to “reflect” such things as he was tuned to receive. The philosophy evolved by such a creature would be expressive of nothing but his own limitations. He would be incapable of understanding the problems that concerned a man. This was also the line taken, more or less, by Dostoevsky and Tolstoi, and it became very popular with artists of all kinds. Wordsworth’s scorn for the botanist became the general attitude towards all men of science. It must be admitted that, judging from biographies of scientific men, there is much to be said for this view. Their favourite authors appear to be Shakespeare and Ella Wheeler Wilcox: they are kind fathers and faithful husbands; in their social relations they are simple-minded snobs; and they are really amused by “lecture-room humour”. It seems unlikely that such people know much of the fierce vitality that sent Saints to rot on pillars and in dungeons, that sent martyrs to the stake, or even that weaker form of vitality that causes our Divorce Court judges to be overworked. That they can understand the universe, when it is obvious they do not understand Clapham, does not seem likely. That, briefly, was the case of the artist against the man of science. The artist was conscious of more things in heaven and earth, staring him in the face, than he believed the man of science had ever dreamt of in his philosophy.

It is evident that the position to-day is rather different. It has become different since the War. It is probable, as we shall see later, that the War itself is partly responsible for the increased attention paid by the artist to science. But the influence was not direct. The artist was not transported with admiration for the men who could make poison-gas,^[1] although he may have been more inclined to believe their philosophy that existence is meaningless. No, the change was, I believe, due to Einstein: in this respect he must be likened to Newton and Darwin. The fact that his theory is completely unintelligible to the enormous majority of those who take an interest in it is not at all to its disadvantage. Rather the contrary. The artist is

attracted by the theory, and respectful to it, not in the least because he understands it, but because he feels it is the result of a most unusual and most powerful *imaginative* effort. It gives him a new conception of the power of the human consciousness. This theory, he is convinced, has come from the heights. It is probable, as a matter of fact, he thinks this because he believes the theory to be about that mathematical platitude, a fourth dimension. The fourth dimension is a phrase to which imaginative people respond with quite extraordinary intensity. Its popularity is like that of giant telescopes, as was proved when a thousand pounds was recently offered for a simple explanation of it. It seems to be the phrase which, to the non-mathematician, is most pregnant with the vast and liberating unknown. If its meaning is ever generally understood, we may anticipate that interest in Einstein's theory will decline. This will be a pity, because the popular reaction to Einstein's theory is perfectly justified. It is the most profound and original scientific theory that has ever been invented, and it displays a kind of imagination almost^[2] unprecedented in the history of science. The feeling of the artist about it is right—it is vastly important to him.

[1] He ought to have been. See *Callinicus*, by J. B. S. Haldane.

Being convinced that the mathematician, at any rate, might be a poet, the respect of imaginative people for science in general has greatly increased. Many of them have decided that science is worth looking into. Unfortunately mathematical physics, the master science of the present day and the one which has furnished ideals for the other sciences, is hopelessly technical. It is agreed that a modern intelligent man, conscious of his responsibilities as an inhabitant of the twentieth century, should be familiar with "the scientific outlook". But to acquire this outlook by brooding over the teachings and implications of modern physics is not easy. Thus although it is the recent astonishing development in physics which is responsible for the renewed public interest in science, it is other sciences that reap the benefit. We have poets and painters who study anthropology and literary critics who read books on the nervous system. The result appears to have been disastrous. At a time when the physicists are abandoning materialism the artists are accepting it. They are accepting, as the last word of science, a picture of the world that belongs to the early bad manner of physics. Again we hear, but this time from our literary men, that slightly hysterical insistence on the duty of intellectual honesty. It must be admitted that they

have been predisposed to accept this view by the War. It is a curious but indisputable psychological fact, perhaps first noted by Tolstoi, that the sight of a large number of naked human bodies makes it difficult to believe that they are animated by immortal spirits possessing an eternal destiny. The sight of the “wastage” that occurred during the War, for those who saw any of it, produced the same curious effect. Also, a psychological fact that cannot be denied, it was difficult to preserve belief in the essential nobility of man when listening to patriotic non-combatants. There can be no doubt that the War, for a large number of those connected with it, has made the acceptance of materialism easier. Even the creative artists, at one time great champions of the spiritual nature of man, are now sufficiently dubious about his nature to be reduced to impotence.

[2] I say “almost” because there was Bernhard Riemann and his disciple W. K. Clifford.

2

The notion that we live in a purposeless universe is so opposed to the mental habits we have inherited that it is a matter of the greatest difficulty to bear it constantly in mind. Most of the people who hold this belief to-day would not do so but for three reasons: the disillusionment caused by the War, their respect for science, and their belief that science preaches materialism. As for the War, that is an experience to which we must accommodate ourselves as best we may. It is consistent with the belief that man is a developing spirit, but it is certainly a proof that he is not very far developed. The respect for science is, I believe, on the whole rather overdone. The respect is a little excessive even when it relates to mathematical physics, but it becomes almost absurd when it relates to some other branches of science. I believe, for instance, that Freud's form of psycho-analysis, some forms of behaviourism, and many of the statements of the eugenists really are as silly as they look. All that they have in common with such first-class mental activities as physics and chemistry is the name "science." It is this name that secures for them such attention as they get from intelligent people who are not cranks. But even physics is a more provisional and more human thing than some romantic references to it would lead one to suppose. Even the tower of the mathematician, which Mr Bernard Shaw imagines to have been always unshaken, has been seriously disturbed on more than one occasion. The student of the history of science will not be too confident even of the "indubitable certainties" of physics when he reflects on the universal passion of belief that attached to the notion of a mechanical ether, for whose present absence from the universe some men of science are still inconsolable, and when he reflects on the fate that has overtaken that "most perfect and perfectly established law", Newton's law of gravitation. There are no indubitable certainties in science, a fact that we who are contemporary with the destruction of the Newtonian system are not likely to forget. There are only provisional hypotheses. It may even be, as Mr J. B. S. Haldane prophesies, that physiology will one day invade and destroy mathematical physics, by which somewhat dark saying I suppose phenomena mathematically may be given up. Whether he means that or not, it is a possibility, as Professor Eddington has hinted. The

scientific practitioner usually treats his hypotheses as tools, but to the layman they become dogmas. One is led to believe this by seeing that many of those who accept materialism on what they suppose to be scientific evidence are rendered acutely unhappy by their belief. A truer knowledge of the status of scientific theories would render this agony unnecessary. There are people with a natural leaning towards materialism, and science, preferably somewhat old-fashioned science, will give them quite sufficient grounds to indulge their propensity with complete intellectual honesty. But science does not, and never has, brought forward sufficient evidence to justify a man turning materialist against his will. And perhaps no man has ever done so. Perhaps one can take the agonies of modern poets too seriously. Many artists, not only small ones, have no real indwelling force such as a man like Beethoven obviously possessed. They are merely very impressionable and *adopt* an attitude towards life, and this attitude is accepted and maintained, not because they really think it is true, but because they derive strength from it. It gives them a centre from which they can work; it gives them a feeling of strength and completeness. The maintenance of their attitude towards life may become the condition that they exist and function as artists at all. Nevertheless, the attitude is maintained only by a constant effort of will, although, since the motive is self-preservation, the artist will nearly always think himself perfectly sincere. But I shall, without going into these refinements, take the unhappiness of our modern literary men at its face-value, those, that is, who believe that the universe is purposeless and think this belief is founded on scientific evidence.

The point of view has been well put recently by Mr I. A. Richards,^[3] a literary critic who thinks it possible that poetry may be destroyed by science. He speaks of the “neutralization of nature” which has been effected by science, and contrasts this with the “magical view” of the world that has hitherto been accepted by artists. What he means by this is that science reveals to us a universe quite indifferent to all human aspirations, whereas artists have hitherto assumed that man is of cosmic significance. The poet must learn to accept the scientific universe and give up believing in things like “inspiration”, “a reality deeper than the reality of science”, and so on. “Experience”, says Mr Richards, “is its own justification”, by which he appears to mean that experience just happens to be what it is by some kind of accident. It points to nothing beyond itself. The ground for this belief is

not, in Mr Richards' case, old-fashioned materialism. "It is not what the universe is made of but how it works, the law it follows, which makes knowledge of it incapable of spurring on our emotional responses." This reminds one of the "iron laws" of the Victorian age, which many people found so depressing, although the logical connection between existence having conditions and existence being purposeless is a little hard to follow. But although the particular iron laws of the Victorians have gone, Mr Richards finds the theory of relativity no more cheering. "A god voluntarily or involuntarily subject to Einstein's General Theory of Relativity does not make an emotional appeal and physics does not find it necessary to mention him." Apparently it is the existence of any law at all that is resented: the poet can feel happy only in a world of pure miracle. I strongly doubt the correctness of Mr Richards' diagnosis.^[4] I am certain that not all poets have been as childish as that. No—the essential element in this general outlook is not that phenomena occur in an orderly way, but that man's existence is not regarded as forming part of some universal purpose. The essential element is the same as in old-fashioned materialism, the "accidental collocations of atoms" theory. The emphasis was on the "accidental" not on the "atoms". This becomes clear when Mr Richards describes the appropriate emotional reaction to his view. "A sense of desolation and uncertainty, of futility, of the baselessness of aspirations, of the vanity of endeavour, and a thirst for a life-giving water which seems suddenly to have failed, are the signs in consciousness of this necessary reorganization of our lives." It is difficult to believe that this state of mind can be produced by the recognition of such facts as that unsupported stones always fall to the ground. But if Mr Richards is right, I suggest that the poets who are so depressed by law and order should study, besides the theory of relativity, Quantum Theory. They will find there much that is, at present, agreeably miraculous. But one need not fly to miracles to get rid of the bug-bear of "unalterable law". It is only necessary to understand the true status of the unalterable laws, and this is just what relativity theory enables us to do.

[3] *Science and Poetry*, 1926.

[4] But possibly Mr Richards means that the scientific description does not include values. See [Section 5](#) of this essay.

3

The idea that there is a conflict between science and art, which is at bottom the idea that there is a conflict between science and mysticism, rests, I have suggested, upon an old-fashioned conception of the status of physics. The first duty of a man who bases his conclusions on science is to make sure that his science is up-to-date. The science that leads to the depressing conclusions I have just sketched is not up-to-date. Until a few years ago the physicist thought that the material universe he dealt with was a real, objectively existing universe in the sense that, in the absence of consciousness, it would be very much the same as it appeared to be. This universe was subject to laws, and these laws might conceivably have been different. There was no *a priori* reason, for instance, why the force of gravitation should not vary as the inverse cube of the distance. There was no *a priori* reason why matter and energy should be conserved. These were laws of governance of the material universe; their discovery had required much effort and the rejection of alternatives. Man was in no sense responsible for them: he happened to live in a universe governed by them. These were the iron laws of the Victorians and are the laws, apparently, that depress modern poets. One of the great discoveries of relativity theory is that these laws need be no more depressing than the laws of Euclidean geometry. No artist has felt his aspirations baseless because he cannot draw a circle whose circumference is six times its radius. He has no more right to despair because there is an inexorable law of gravitation. This has been made clear by Professor Eddington, whose mathematical development of relativity theory is of great philosophical importance, and would, in a more adequately educated community, be given more newspaper headlines than Tutankhamen. The real universe, according to relativity theory, is a four-dimensional world of point-events. Of the nature of point-events we know nothing. All that we require to know, for the purposes of physics, is that it takes four numbers to specify a point-event uniquely, and that some kind of structure—a minimum amount of structure—may be postulated of the world of point-events. We then find, purely by mathematical processes, that certain characteristics of this world will have the quality of permanence. The mind, faced with this world of evanescent point-events, selects those

characteristics that are permanent as being of special interest. This is merely because the mind happens to be that kind of thing. As a consequence of this predilection of the mind there arises space and time, matter, and the laws of nature. There arises, in fact, the “objective universe”. The real world of point-events has many other characteristics to which the mind pays no attention. A different principle of selection, exercised on the same total world of point-events, would result in an utterly different universe, a universe that is, for us, quite unimaginable. And the universe that the mind has selected and constructed from the world of point-events does not in the least depend on what the point-events *are*. All that is necessary is that a certain minimum amount of structure should be attributed to the world of point-events. It is from the relations between the point-events, quite independent of their substance, that the mind has created the material universe and its laws. These laws, it must be emphasized, are *necessary* consequences of the mind’s selective action. They are necessary in the same sense that the sum of the three interior angles of a Euclidean triangle must be two right angles. Of the underlying reality deduced by physics we can say almost nothing. It may be what Newton called the “sensorium” of God, and the point-events may be his thoughts. They do not succeed one another in time for, at this stage of analysis, space and time are “merged in one”. This perfectly gratuitous hypothesis may appeal to some mystics, for our thoughts, considered as belonging to the world of point-events, would be part of the thoughts of God. It would be indeed true that in him we lived and moved and had our being. We see, then, the limitations of physics. All that depends on the *structure* of reality belongs to physics, including other universes than ours. All that depends upon the *substance* of reality for ever lies outside physics. As to the actual universe we live in, why we should regard it as actual is a problem for psychology. The difference between the actual and the non-actual is a distinction conferred by our minds. It is very probable that the whole movement of the universe in time is also contributed by our minds. It seems to be true that events do not take place—we come across them. Why we do not know the future is again a question of psychology. Ignorance of the future, like the existence of the material universe, is a clue to the constitution of our minds. This has a bearing on the question of “purpose” in the universe. The conception of purpose seems to suppose a process in time, and therefore may be a totally irrelevant idea when applied to reality.

The philosophical implications of relativity theory will doubtless take a long time to work out. The four-dimensional universe of point-events is something that can be argued about but it is, to use an old-fashioned phrase, “inconceivable”. Mankind, excepting professional logicians, never remains content with the inconceivable. A purely logical conclusion is not enough; it has to be grasped imaginatively, by which I do not necessarily mean that it has to be pictured. To become familiar with a theory does not merely mean that one is able, as a form of mental wire-walking, to slip nimbly back and forth over the logical connections of the structure. It means taking it into oneself in some indefinable manner—becoming “intimate” with it. Only when a theory is “realized”, as we say, do we feel that we truly understand it. Ideas, points of view, that we were able to see only in flashes, become part of our normal intellectual equipment. The process may well be called a growth of consciousness. There are ideas which our consciousness, when it first approaches them is, as it were, too flabby to grasp. We first have to exercise our mental muscles. Every student of a line of thought such as mathematics, which is rather outside our normal preoccupations, becomes aware of an actual change in his mental powers. Notions so abstract that at first they seemed almost meaningless gradually become perfectly clear and permanent additions to one’s mental resources. Students of musical composition find that their capacity for mentally hearing a number of parts rapidly increases. In some cases it is almost as if a new faculty of the mind were born and developed.

The physics of recent years has made heavy demands upon our capacity for realization. The electron theory, with its analysis of matter into “disembodied charges of electricity” required, for its understanding, the breaking up of old habits of thought. To young students the idea was, at first, extremely baffling—almost nonsense. To realize it one had to make more abstract one’s idea of matter until the notion of “substance” was replaced by the notion of “behaviour”. Anything that behaved in the way characteristic of matter was matter. The central idea of the restricted principle of relativity, the idea of different time-systems, was still more difficult to grasp. In this case we had to become convinced that our ordinary idea of simultaneity, an idea which seemed perfectly clear, was really a bogus idea. The attacks on the theory of relativity show, for the most part, merely that their authors are unable to abandon old habits of thought. With the complete theory of relativity, as we have it now, the task of adjustment

has become enormous. There cannot be, even now, more than very few scientific men who naturally approach a problem from the point of view of relativity theory. In most cases a conscious effort of mental preparation is required, such as occurs when a novelist, sitting down to continue his work, deliberately thinks himself into the appropriate frame of mind. Yet doubtless the next generation or so will think in terms of relativity theory as naturally as we thought in terms of the Newtonian system. I would not hold it as impossible that the human mind may come to realize, imaginatively as well as logically, the four-dimensional space-time continuum. But it seems that the mind of the physicist, at any rate, will have to do more than become familiar with relativity theory. It will have to accommodate itself somehow to the quantum theory for, although we can write down the laws which govern sub-atomic phenomena and make deductions from them, these laws are, at present, unintelligible. An electron behaves as if it had foreknowledge of what it was about to do and could make the mathematical calculations necessary to achieve its end. We cannot admit this to be possible, and we can only suppose that the difficulty arises from the way we think about things. We must learn to think in a different way, and what the consequences of that new way of thinking will be no one can say. We know very little of the possibilities of the development of the human consciousness.

The proper attitude to-day in which the problem of man's place in nature should be approached is one of bewilderment and humility. Both the material universe and the mind of man are very mysterious things. At the present time it is only an inadequate mind which is confident that it knows what is impossible. There was never a time when hearty dogmatism and loud confidence were more out of place. We must think as best we can, of course. The next step upward in the development of the human consciousness will not be achieved by either slovenly credulity or slovenly scepticism, but only by a terrifying mental travail. I see a human mind as some multiple plant, here in full flower, there still in the bud. Different minds have flowered in different ways. Beethoven's *Heiliger Dankgesang eines Genesenen an die Gottheit* points to the complete development in him of something which those of us who understand him have only in embryo. In those who do not understand him it is non-existent. And the great mystics ought at least to make us doubt whether it is we who are not deficient rather than they who are mad. It is rash to dismiss our exceptional moods, our

strange flashes of what seems like insight, as mere whimsies without significance. They may be faint stirrings of the next thing that is destined to become fully alive. All that we can say is that the mind lives in a universe largely of its own creation, and that the universe, together with the mind, will change in ways we cannot foresee.

4

We have seen that the philosophy that regards man as a meaningless accident in an alien universe receives no support from modern physics. The true ground of that philosophy is now, as it always has been, the apparently meaningless misery that forms part of life. It is not by mistaking matter for an ultimate reality or by pondering on the fact that laws of nature exist that we can conclude that man is of no cosmic significance. That conclusion can be reached logically only on the basis of arbitrary assumptions. But the conclusion is not, in fact, reached in that way: it is reached through feeling. And it cannot be transcended by a logical process, but only in virtue of a mystic experience.

The old materialistic outlook, although it no longer has any scientific justification, is still active in many branches of science. It has made popular certain types of explanation and is the cause of the direction pursued by certain researches. In particular it has led to a great deal of useless or misleading work being done in the attempt to reduce qualitative to quantitative differences.

A good deal of what passes for scientific work amongst eugenists and psychologists consists of attempts to match things which are qualitatively different. This is the favourite procedure of that kind of psycho-analysis which reduces everything to sex. Discrimination is fatiguing; also, it makes appeal to sensibilities which many earnest "scientific workers" do not possess. It is much easier to make measurements than to know exactly what you are measuring.

To give up the ideal of measurability would be equivalent, to many people, to abandoning "science" altogether. "Science is measurement", we are informed. This ideal is borrowed from physics, the science whose aim it is to give mathematical descriptions of phenomena. But we may have branches of knowledge that may fairly be called science although they are not mathematical. We may find it necessary to use concepts that cannot be mathematically defined. It may not be mere lack of knowledge which prevents biology, for instance, from being a mathematical science. It may

be impossible in the nature of things ever to give the equation to a chicken. But the bias towards measurability is very strong and has led to measurements being made, particularly in psychology, where we really have no clear idea at all as to what is being measured. When, for instance, Professor Karl Pearson compares fraternal resemblances in such things as stature and arm-length with fraternal resemblances in intelligence and conscientiousness, what exactly is he doing? A great deal of what is called experimental psychology impresses one as being nothing but the application of an inappropriate technique by exceptionally innocent and unworldly “scientists”. The methods found so successful in physics are applied to everything under the sun. It is pretty obvious that this is not due to some mystic, Pythagorean conviction that number is the principle of all things, but merely to mental inertia. Many “intelligence tests” and many of the statistical results obtained by the eugenists impress the ordinary person as being laughably superficial. In their eagerness to “measure” something our researcher seem to lose their ordinary common sense, whereas their subject really requires the subtlety and sympathy of a very good novelist. It is amazing the number of dull, unimaginative people who find a congenial life work in prosecuting researches in pseudo-science. The ordinary public, unfortunately, does not discriminate between one kind of science and another, with the result that the contempt they rightly feel for some so-called men of science is apt to be extended to all scientific men. Thus Mr G. K. Chesterton, having heard that some “scientists” explain the shape of a church spire as symbolical of phallic worship, begins to doubt the whole Royal Society. It must be remembered that in science real insight and imagination are as rare as in any other human activity. In the clear-cut sciences, such as physics and chemistry, where the right way of attacking problems is known and where an elaborate technique has been built up, there is plenty of room for valuable routine work. All the difficult preliminary work of getting right conceptions and principles has been done. The routine worker can measure the electric capacities of different condensers because the difficult notion of electric capacity has been made clear by his masters. But the routine worker in psychology who measures “intelligence” is not doing anything definite at all. His subject is not yet ripe for the application of such exact methods. In this way the prestige of physics has exerted a harmful influence on the study of psychology. It is true that some experimental psychologists are becoming aware of the fact

that they do not always know what they are measuring. There are controversies as to what a given set of measurements has measured, and some measurements seem to be undertaken on the off-chance that a meaning will some day be found for them. It is not suggested that all experimental psychology is of this kind, but it is certainly true that many psychological papers, complete with correlation coefficients and “curves” of all kinds, wear an air of precision to which they have no real claim.

A more definitely materialistic bias is observable in the attempts to explain psychological happenings in terms of physiology. The result is that learned and acute men, caught in the jungle of neurology, painfully fight their way out with some such epoch-making discovery as that one learns a subject more rapidly if one is interested in it. This result, which is supposed to be incompatible with the purely physiological theory of the mind, owes all its difficulty to that in compatibility. Otherwise it is a perfectly obvious fact of experience. If it were not for the prestige achieved by materialism in the Victorian age it is probable that psychology would be very much further advanced than it is. But the side-tracking influence of that philosophy has meant that psychologists have had painfully to discover the obvious. But if materialism, in small doses, delays the recognition of the obvious, it does, when fully developed, deny the obvious. This is what the behaviourists do. They deny that we think or that we can form images in our minds. The only possible answer to this theory is a satire, as when Voltaire answered the theory that in this world everything is for the best in the best of all possible worlds by writing *Candide*. But in this queer modern world behaviourism, instead of being greeted with laughter, is answered carefully and politely, apparently in the spirit in which Monsieur Bergeret shook hands with the *vers libriste* poet, “for fear of wronging beauty in disguise”. The position of the ordinary man in face of these theories is, nevertheless, a difficult one. Behaviourism may sound to him nonsense, but so does non-Euclidean geometry. His natural reaction would be to class both of them with the theory that the English are descended from the lost ten tribes of Israel. Nevertheless, non-Euclidean geometry is not nonsense. In these circumstances it is probably fortunate that there are people patient enough to prepare careful and reasoned refutations of any whimsy that anybody cares to put forth. The extraordinary predisposition of the learned towards concocting merely silly theories must always be borne in mind. Studious persons often have a very small range of experience of life; they have

nothing like so broadly based a sense of probability as the ordinary man of the world possesses, which is why so many of them seem curiously innocent and gullible. The beaming and genial professor expounding his theory often seems curiously like a child playing with toys. The mixture of amusement and respect with which the world watches him is, on the whole, the correct reaction. As long as he is dealing with the incomprehensible one may grant him authority. Nobody dreams of questioning astronomical pronouncements about forthcoming eclipses. But when he is talking about the very stuff of our ordinary experience, as in psychology, we do wrong to accept the obviously absurd for fear that it cannot be as silly as it looks. A great deal of what is called psycho-analysis, for instance, is merely silly. Only people singularly deficient in common-sense and completely lacking in a sense of humour could have invented anything so preposterous. Undoubtedly some pathological states are of sexual origin, but the lengths to which the theory has been carried and the kind of interpretations that are given make the development of psycho-analysis one of the greatest psychological curiosities of our time. Whole-hearted belief in psycho-analysis certainly points to the existence of a complex. As with any other complex, it is defended by arguments to which none except those who are similarly afflicted can attach the slightest validity. The complex is strongly materialistic, not in the sense that everything is reduced to “matter and motion”, but in the sense that the lowest human activities are made explanatory of all the rest. One often finds, associated with a belief in materialism, a desire to deny any form of spiritual excellence. The ostensible motive is simplification, as when material substances are reduced to a small number of chemical elements; but it is usually obvious, from the forced explanations that are attempted, that the real motive is something very different. Much, of course, must be attributed to insensitiveness, as we see when we turn to psycho-analytic explanations of works of art. The extraordinary force of the psycho-analysts’ complex is well shown by the sort of arguments they find convincing. Thus they may profess to show that artistic tastes never exist without suppressed sexual desires. Their way of establishing this fact, which is chiefly by asserting it, is comparatively rational. But they then proceed to the statement that a taste for art is merely a disguised form of sexual desire. They might as well say that it is a disguised form of hunger, since artists are quite as notorious for being

hungry as for being erotic, and artistic tastes are never found to exist in a man who takes no nourishment.

Not only much modern psychology, but some other modern sciences such as comparative religion, are prone to a certain fallacy that may be called the fallacy of “explanation by origins”. This kind of explanation has been made popular by the theory of evolution, and the fallacy consists in supposing that to give the historical antecedents of a thing is to give an analysis of that thing. Thus, some authorities suppose that by showing that religion has developed from primitive magic rites, they have thereby proved that religion is nothing but a disguised form of magic. One might as well say that an oak-tree is a disguised form of an acorn, or that a man is a disguised form of an amoeba. But this error is too glaring to be committed by more than a small percentage of our modern “thinkers”. A much more insidious danger is that this type of explanation leads one to under-estimate the complexity of the thing to be explained. There is a tendency to neglect those factors in the final product which cannot be traced in its historical antecedents. This is one form of the widespread error of undue simplification. No human mind can deal exhaustively with concrete facts. Every natural entity, whether it be a flower or a nation, contains far too many factors for thought to grasp it completely. The art of human thinking is to make useful abstractions. Any man is a very complicated creature. All the artists and scientists of the world could not describe him exhaustively. But for the purposes of war every man under a certain military rank was regarded as a physical structure supporting weapons and a stomach on two legs. This abstraction was useful for the purposes for which it was invented. A somewhat different abstraction is required when a man is considered as a voter. When a man is considered as a “hand” or a “worker” it is found that slightly more complicated abstractions are required. In fact, the great fault of economic theory has been that its “economic man” was too simple an abstraction. The economist left out certain factors in his conception of man, with the result that his plans, when applied to real men, do not work. I am suggesting that the sciences which ape physics suffer, amongst other things, from inadequate abstractions. This is not surprising, for there is every reason to suppose that the extraordinary difficulties experienced by physics itself, at the present day, are due to the same cause. An analysis of this position will show us the direction of the probable future development of science and help us to see in what consists the importance of the arts.

Many people, including some scientific men, take science too seriously. They think that science gives a far more comprehensive picture of reality than it really does. There have been philosophers who have gone so far as to suppose that those factors of experience that science does not find it necessary to talk about do not really exist. This is the basis of the belief that colours, sounds, and scents have no “objective” existence; they exist only in the mind, whereas such qualities as mass and extension are supposed to exist independently of the mind. It is true that science does not find it necessary to refer to colours, sounds, and scents in giving its description of nature, whereas it does find it necessary to refer to mass and extension. But that does not prove that the former qualities are not as real as the latter, are not as indubitably part of the universe. The scientific concepts have by no means proved themselves adequate to account for the whole of experience. Nearly everything of real importance to man lies at present outside science. The fact is that science was undertaken as an intellectual adventure: it was an attempt to find out how far nature could be described in mathematical terms. Certain primary conceptions—time, space, mass, force, and so on—all of which can be defined mathematically, were adopted, and it became a highly absorbing game to find out how much of what goes on around us could be described, mathematically, in terms of these conceptions. The success of this effort has been so astonishing that some scientific men have forgotten to be astonished. They have come to take it for granted that a complete mathematical description of the world should be possible. This assumption is not a rational one: it is a pure act of faith. The great founders of the scheme made no such mistake: they were quite aware of the precarious nature of their enterprise. Thus, Newton, the greatest and most successful of them all, says that, if they find the mathematical method does not work, they must try a different method. The mathematical method, which is the very essence of modern science, has, however, worked splendidly. From the time of its origination in the seventeenth century until the present day it has had no serious rival. The ancient æsthetic principle, which led to the conclusion that the planets moved in circles because the circle is the only perfect figure, is still used by

theosophists, but not by men of science. Similarly the old moralistic principle, which explained the fact of water rising in a pump by saying that nature abhorred a vacuum, possibly lingers on only in such superstitions as that sunlight puts the fire out. In more modern times the only notorious rival of the Newtonian method was the dialectic method of Hegel, who evolved the laws of the universe from his inner consciousness. But the best-known result of this method, that there could not be more planets than were known to exist, happened to be published on the very day that a new planet was discovered. The mathematical method, then, is at the present day without a rival. But, although we cannot at present imagine what could replace the mathematical method, we must be careful not to exaggerate the significance of the results that have been achieved by it. For these results depend not only on the method, but also on the material the method has to work with. And there is good reason to suppose, in the present state of physics, that the material with which science has worked hitherto is turning out to be not quite satisfactory.

This material is chiefly the Newtonian set of abstractions. Newton postulated, as the fundamental constituents out of which the perceived universe is built up, Space, Time, and Matter. Space and time he regarded as absolute and as quite independent of matter. Matter was an enduring substance that simply inhabited space and time. The analysis of these conceptions has resulted in the Einstein theory, in which neither space, time, nor matter are fundamental. The interesting thing about this analysis, from our present point of view, is that it shows clearly what arbitrary elements are present in the scientific description of the universe. For we must remember that moral and æsthetic elements were ruled out of the real universe simply because science did not find it necessary to mention them. The foundation stones of the scientific edifice, namely space, time, and matter, were supposed to be the only realities. Everything else was a sort of illusion. Men who must have been theory-mad soberly maintained that little particles of matter wandering about purposelessly in space and time produced our minds, our hopes, and fears, the scent of the rose, the colours of the sunset, the songs of the birds, and our knowledge of the little particles themselves. The sole realities were the little wandering particles and the space and time they wandered in. The existence of everything else depended on the mind, and was inconceivable without the mind. It is interesting, therefore, that science has now reached a position where space,

time, and matter also depend on the mind. In giving a scientific description of the universe Einstein does not find it necessary to begin with space, time, and matter. These entities become “derivative”. The universe becomes more spectral than ever if we are going to adopt the materialist principle that what depends on the mind does not really exist. Even the universe of wandering particles is comparatively cosy compared with this modern universe of undefinable “point-events”. But if we do not adopt the materialist principle we may assert that moral and æsthetic values are as much a part of the real universe as anything else, and that the reason why science does not find it necessary to mention them is not because they are not there but because science is a game played according to certain rules, and those rules have excluded these values from the outset. The life-insurance actuary may, for his purposes, neglect many things about men, and yet calculate, quite correctly, what percentage of them will die at forty. But he has not proved that the qualities he has neglected do not exist simply because they do not come in to upset his calculations. A politician finds that he has to base his calculations on quite different aspects of mankind from those found satisfactory by the actuary. In the same way, a mountain is a different thing to a poet from what it is to a man of science. For the kind of understanding of the universe that the man of science is after, the mountain is merely a heap of certain kinds of matter weighing so many millions of tons. The poet, who is after a different kind of vision, finds it necessary to take into account quite other factors which enter into his total experience of the mountain. The scientist may also experience emotions of awe and reverence in the presence of the mountain, but for the purposes of his science these factors of his experience may be neglected. He *abstracts* from the total concrete fact of his experience of the mountain. The mountain, as he describes it in the scientific paper he proceeds to write, is a mere pale shadow of the real mountain; he probably leaves it indistinguishable from any other mountain that happens to weigh the same, just as to the life-insurance actuary all men of forty are exactly alike. If we believe that the factors in experience that the scientific man neglects are quite as real as those he takes into account, it becomes a matter for wonder that science is possible. How is it that science forms a closed system—that nothing from the worlds it neglects ever comes in to disturb it?

It is one of the great services of relativity theory to philosophy that it provides an answer to this question. The answer is that the entities

discussed by physics are defined in terms of one another. The three hundred years of building up exact science really amounts, in the last analysis, to doing what the dictionary compiler did when he defined a violin as a small violoncello and a violoncello as a large violin. Of course, if this statement were literally true, science would give us no information about the universe at all. Nevertheless, the statement is true about the actual procedure of science, and it is in virtue of this procedure that science forms a closed system. But what is left out of this description is the scientist himself. The mysterious process which is not taken into account in this description of the scientific method is the process by which the consciousness of the scientist makes contact with the entities he is talking about. In deducting the world from “point-events”, for instance, we begin by talking about something we have no direct cognisance of, namely point-events. From point-events we deduce “potentials”—again a mere word. But from potentials we deduce “matter”, and here we are talking of something of which we have direct knowledge. Similarly, the circular definition of violin and violoncello tells us nothing as it stands. But to a man who can identify one of these entities, to a man who has ever seen a violin, it gives genuine information.

We need not be surprised, therefore, that nothing from the outside ever seems to disturb the equanimity that reigns within the closed system of physics. The abstractions with which it begins are all it ever has to deal with. There are no subsequent fresh contacts with reality. If the region covered by relativity theory embraced the whole of physics it would seem that, so far as physical science is concerned, we knew all that there is to be known. But it is notorious that, of recent years, an entirely new set of phenomena has been discovered in physical science. These phenomena arise when we consider, not matter in bulk but matter in its smallest particles. These phenomena are, at present, strictly incomprehensible. The celebrated quantum theory provides us with rules for dealing with some of them, but does not make them intelligible. It seems that science has here reached its limits. Professor Eddington has even hinted that these phenomena may indicate that the universe is finally irrational, that is, that the attempt to describe nature mathematically will have to be given up. This is a possibility that Newton foresaw. But it seems more likely that our present state of bewilderment has a different cause. That cause, we shall probably find, is the insufficiency of the abstractions hitherto used in science. We have to go back to the concrete facts of experience and build up

a richer, fuller set of abstractions. Physics is now paying the penalty of inadequate abstraction. In particular, it must revise its notions of space, time, and substance. This revision is quite independent of the Einstein theory, and is made necessary, not by that theory but by the quantum theory. A first attempt at this revision has been made by that great mathematical philosopher, Professor Whitehead.^[5] We need not deal with his investigation, which is at present in a highly technical state. The space and time of the new theory are interconnected and do not consist of independent volumes and instants. Every volume of space has reference to the whole of space, and every moment of time refers both to the past and the future. Hence both memory and expectation are given a rational basis. On the old view, as Hume pointed out, there is no reason whatever to suppose that the order of nature should continue. Why do we expect that the force of gravity will be in existence to-morrow? There was no *reason* at all for this expectation or for any other. That is to say, the whole of science itself was based on blind faith. The new foundations of science make science itself a rational activity. As for the notion of “substance”, Professor Whitehead proposes to replace it by the notion of “organism”. We may imagine an electron, for instance, as a repeated pattern of events. One of the great difficulties of the quantum phenomena is that an electron seems to pass from one place to another without passing through the intervening space. On the basis of the new abstractions this difficulty can be overcome. We have to imagine an electron as requiring a certain time to manifest itself—just as a tune does.

[5] *Science and the Modern World*.

From our present point of view, however, the chief interest attaching to these new foundations for science is the place occupied in them by the intuitions of the poets. Mr Richards, literary critic, tells us that the poets must learn from science; Professor Whitehead, mathematician and physicist, tells us that science must learn from the poets. Instead of the poet having to realize that his intuitions are illusory and belong to a childish, *démodé* view of the world, it is the scientific man who must realize that his abstractions are too thin and narrow to be any longer useful, and that the poet makes closer contact with reality. When Wordsworth says:

“Ye Presences of Nature in the sky
And on the earth! Ye Visions of the hills!

And Souls of lonely places! can I think
A vulgar hope was yours when ye employed
Such ministry, when ye through many a year
Haunting me thus among my boyish sports,
On caves and trees, upon the woods and hills,
Impressed upon all forms the characters
Of danger or desire; and thus did make
The surface of the universal earth
With triumph and delight, with hope and fear
Work like a sea?..."

he is not, according to Professor Whitehead, expressing fantasies that the strong-minded realist can afford to neglect: he is describing the actual concrete facts of experience, facts which, says Professor Whitehead, "are distorted in the scientific analysis". It is the artist not the scientist who deals most adequately with reality. It is the man of science, taking his pale abstractions for the only realities, who dwells in dream-land.

So far as we can see at present, however, science cannot abandon its method. It cannot deal with the whole concrete fact: it must continue to make abstractions. But the present *impasse* in scientific theory is an indication that it must go back to the beginning and include more factors of the concrete fact in its abstractions. It seems likely that, in doing so, it will have to presuppose a philosophy very different from the materialism hitherto current amongst scientific men. The world will have to be regarded as an evolutionary process, where "patterns of value" emerge. It will have to be regarded as an interconnected whole, and the separation of mind from matter, and mind from mind, will have to be replaced by a conception which regards these distinctions, in their present form, as unreal. One very desirable result of this transformation will be that the arts will be taken seriously. The old outlook did not regard values as inherent in reality. They were merely expressive of the accidental human constitution, but had no cosmic significance. Art existed to provide a unique thrill, called the "aesthetic emotion". On the new outlook the function of the arts is to communicate knowledge and, moreover, the most valuable kind of knowledge. Art, much more than science, expresses the concrete facts of experience in their actuality. Music, in particular, finds its highest function in revealing to us the possibilities of the spirit of man himself. The music of

such a man as Beethoven is a revelation of existence from the vantage point of a higher consciousness. It is, we may hope, prophetic of the future development of the race. Not only art, but morals, acquire vastly greater importance on the new outlook. Morals are no longer a purely private concern, expressive of a particular human constitution in an alien, strictly non-moral universe. Men are no longer justified in believing that their only duty is to preserve their self-respect and to make the most of their opportunities.

Science, in view of our increased knowledge of its aims and powers, can no longer be presented to us as a tyrant. Science assumes certain fundamental principles and entities, and there is an arbitrary element in these assumptions. What science does not assume does not thereby not exist. It gives, and it appears that it must forever give, a *partial* description of the universe. The fact that the elements of reality it leaves out do not come in to disturb it is no presumption against the existence of these elements. For science forms a closed system simply because it employs the device of cyclic definition. The teachings of science, so far as the spiritual problems of men are concerned, need no longer be regarded as stultifying: they are merely irrelevant.

Transcriber's Notes:

A List of Sections has been provided for the convenience of the reader, and is granted to the public domain.

Punctuation and spelling inaccuracies were silently corrected.

Archaic and variable spelling has been preserved.

*** END OF THE PROJECT GUTENBERG EBOOK GALLIO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.